

# Suvidha सुविधा

Live lecture translation that  
*refuses to translate the vocabulary.*

[professortranslate.onrender.com](https://professortranslate.onrender.com)   [github.com/kingkamra1729/ProfessorTranslate](https://github.com/kingkamra1729/ProfessorTranslate)

The professor says: “So the eigenvalue of this matrix tells us how much the eigenvector is stretched.”

तो इस **matrix** का **eigenvalue** बताता है कि **eigenvector** कितना

|                |          |        |         |        |         |        |
|----------------|----------|--------|---------|--------|---------|--------|
| हिन्दी         | ENGLISH  | हिन्दी | ENGLISH | हिन्दी | ENGLISH | हिन्दी |
| <b>stretch</b> | होता है। |        |         |        |         |        |
| ENGLISH        | हिन्दी   |        |         |        |         |        |

One sentence, two voices, in the student’s earpiece. The explanation is translated. The vocabulary never is.

WHERE THIS CAME FROM

I did not start from a dataset. I started from the people sitting around me.

My college takes students from every part of the country, and we arrive speaking different mother tongues into the same lecture hall, where everything is delivered in English. I watched classmates who had cleared the same entrance exam I had — who understood the subject perfectly well — fall behind because of the language it arrived in. They would ask me afterwards what the professor had said. Not what a word meant. What the *sentence* meant.

Then I saw the live interpretation used in parliament, where a speech reaches every member in their own language as it is being given, and the question became obvious: why does a lecture hall not have this?

## Any student sitting in a lecture delivered in someone else's language

I saw this in India, but nothing about the problem is Indian. It is the exchange student in Berlin, the refugee finishing a degree in a country they arrived in last year, the engineering student anywhere whose textbooks are in English and whose thinking is not.

Wherever education is delivered in a language of prestige rather than a language of home, the same students fall behind for the same reason — and it is never because the material was beyond them.

The system is built language-agnostically. It runs today in English, Hindi, Bengali and French; adding another is one entry in a config file plus a handful of example sentences to pin the register.

### THE BARRIER

## Not the terminology. The explanation.

A student can look up *eigenvalue*. What they cannot look up is the sentence the lecturer improvised to make it land — “*so it tells you how much the vector gets stretched*” — the aside that connected it to last week, the “*think about why that must be true.*” That sentence is never written down. It exists for four seconds and is gone.

And the obvious fix makes it worse:

तो इस आव्यूह का अभिलक्षणिक मान...

What an ordinary translator produces. Correct, literary, and a word that appears in no textbook, no exam, and no job interview.

The student now understands the sentence and cannot answer the question. The sentence got easier; the degree got harder. We are not trying to spare anyone from learning English — we are trying to stop the language from standing between them and the subject while they do.

### HOW IT WORKS

## Four decisions, each one measurable

### The model never sees the terms

Each is replaced by a sentinel before translation and spliced back after. It is not asked nicely to preserve them — it is never given them. Measured at 0% loss across 162 translations.

### The same pass decides the voice

Restoring a term is the only moment we know which characters are vocabulary and which are explanation. That boundary drives both the highlight and the voice switch, so the subtitle and the audio always agree.

## It speaks the way people speak

Real classroom speech code-mixes: अब देखते हैं कि **determinant** **zero** हो तो क्या होता है। A translation that sends a student to a dictionary has failed at its one job.

EVERYTHING IT DOES

## What runs today, and what is designed but not yet built

WORKING NOW

- **Live translation to an earpiece.** The professor speaks; each student hears their own language, seconds behind.
- **Technical terms never translated** — in the subtitles and in the audio, spoken by a second voice.
- **Every student picks their own language,** switchable mid-sentence, in one shared lecture.
- **Subtitles on screen** with protected vocabulary marked, alongside what the professor actually said.
- **Diagrams from the lecture itself.** Wolfram renders the plot a lecturer is describing — published only once the professor approves it.
- **Spoken descriptions of every diagram,** translated the same way, so a student who cannot see the board hears what it shows.
- **Recorded lectures,** archived as text rather than audio — so they replay in a language nobody chose that day, at any speed, or through a screen reader.
- **Course files build the glossary.** Upload a PDF of last year's notes, or point it at a syllabus URL.
- **A pre-flight check** that tells a professor whether the room's hardware can actually do this, before the class arrives.

## It catches its own gaps

The glossary is the single point of failure, so the system listens for vocabulary it does not yet protect and offers it to the professor mid-lecture. Accepting one protects it for the rest of the class.

DESIGNED, NOT YET BUILT

- **Animated demonstrations.** Diagrams are static plots today. A wave that actually oscillates and a vector that actually rotates is the next thing to build.
- **Generated video** for concepts that need more than a plot, for students following on a computer rather than a phone.
- **Definitions preserved like terms.** Right now only the vocabulary is protected; a spoken definition is still translated, and it should not be.
- **Extraction from photographs of handwritten notes,** so a glossary can be built from what is actually on the board.
- **More languages,** and instruction in languages other than English.
- **Cloud speech recognition** as an option for noisy halls and strong accents, behind the same interface as the browser one.
- **Recordings that survive a redeploy,** which today they do not.

## Everything in the left column runs. Nothing is mocked.

| MEASURED                                           | RESULT   |
|----------------------------------------------------|----------|
| Technical terms lost in translation, over 162 runs | 0%       |
| Words lost from continuous fragmented speech       | 0 of 222 |
| First translated word on the student’s screen      | ~600 ms  |
| Automated assertions, across eight suites          | 100+     |

**This is a prototype, not a product.** It was built in a week and has not yet faced a real class. The pipeline is measured and the numbers above are real, but speech recognition on accented English in a noisy hall is the weakest link, and a mistranslated sentence costs a student more than no translation would.

**The one thing we did not build is speech.** Recognition and synthesis run in the browser — free, keyless — because a lecturer should be able to walk into a hall and use this without anyone having provisioned anything. A device with no voice installed for a language produces silence; the app detects that and says so rather than failing quietly. Open `/check` on any device to see.